

Cloud Computing

BCA 5th Semester, Batch BCA2024
School of Engineering

Presented by: Udaya Joisa

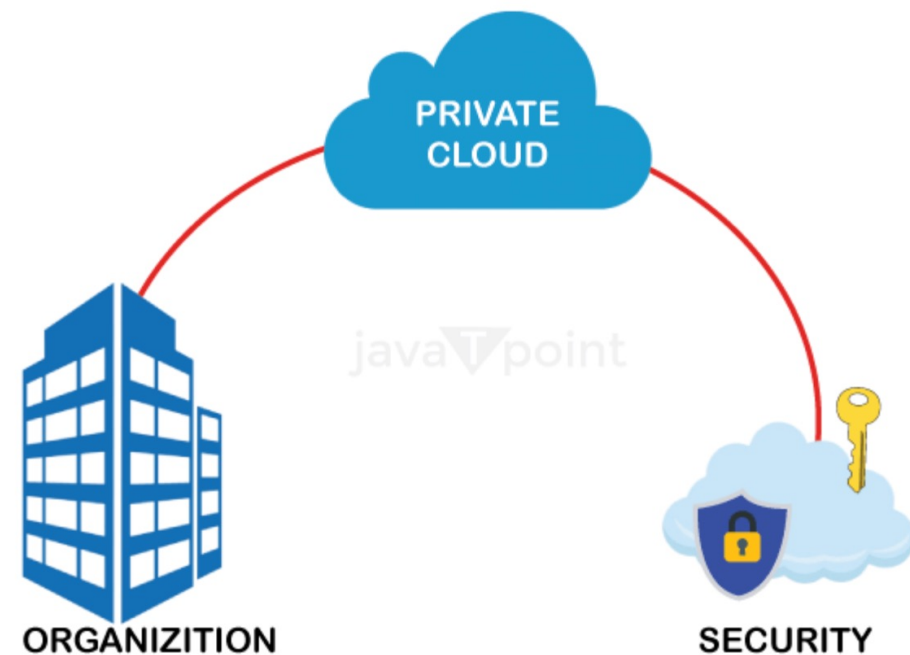
Major Cloud Deployment Models

- **Public Clouds:** Public Cloud are the most common deployment models in which necessary IT infrastructure is established by a 3rd party service provider who makes it available to any consumer on subscription basis.
- **Accessibility:** Accessible to anyone with an internet connection.
- **Shared Infrastructure:** Resources are shared among multiple users.
- **Scalability:** Easily scalable based on demand.
- **Pay-per-Usage:** Users pay only for the resources they use.
- **Managed by Service Providers:** Maintenance and security are handled by the service provider.
- **Reliability and Redundancy:** Providers ensure high reliability with redundant systems.



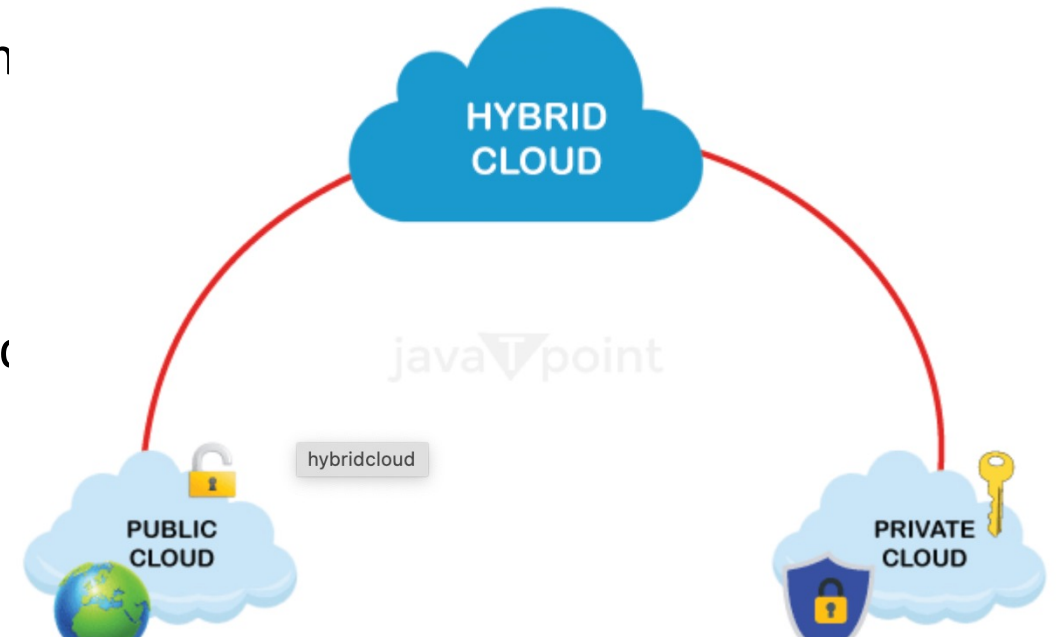
Major Cloud Deployment Models

- **Private Clouds:** A private cloud is a cloud computing environment that's exclusively used by a single organization. It's also known as a corporate cloud or internal private cloud.
- This is driven by the need of keeping confidential information within the organization's control or to make use of owned infrastructure.
- **Exclusive Use:** Dedicated to a single organization.
- **Control and Security:** Higher control over data and security.
- **Customization:** Can be tailored to specific needs.
- **Scalability:** Can scale within the organization's resources.
- **Performance and Reliability:** Better performance control.
- **Compliance:** Easier to meet regulatory requirements.



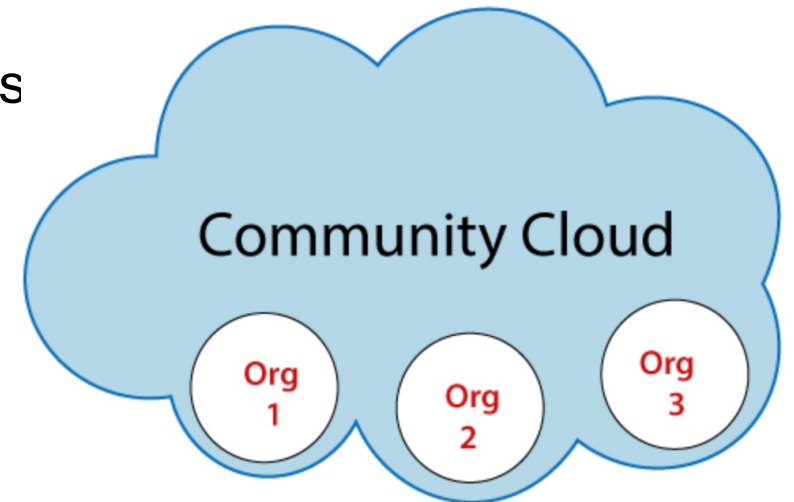
Major Cloud Deployment Models

- **Hybrid Clouds:** Hybrid clouds are composed of public cloud resources and privately owned infrastructure, allowing data and applications to be shared between them.
 - These are created when private cloud resources are unable to meet user's requirements and organisation's needs.
 - **Integration of Public and Private Clouds:** Combines the benefits of both.
 - **Flexibility and Scalability:** Dynamic resource allocation
 - **Enhanced Security:** Control over sensitive data.
 - **Cost Optimization:** Balances cost and efficiency.
 - **Data Portability:** Move data between clouds as needed



Major Cloud Deployment Models

- **Community Clouds:** Community cloud is shared among several organizations with common interests (e.g. research institutes, security etc)
- **Shared Infrastructure:** Accessible to a specific community.
- **Community-Specific Services:** Tailored to meet the community's needs.
- **Community Ownership and Management:** Managed by one or more organizations within the community.
- **Enhanced Security:** Higher security than public cloud.
- **Cost Sharing:** Costs are shared among the community members.



Major Cloud Deployment Models

- **Multi-Cloud:** Multi-cloud uses multiple cloud services from different providers to meet various needs.
- **Multiple Cloud Providers:** Utilizes services from various providers.
- **Diversification and Risk Reduction:** Reduces risk of vendor lock-in.
- **Flexibility and Vendor Independence:** Choose the best services from different providers.
- **Service and Cost Optimization:** Optimize costs by selecting suitable providers.
- **Enhanced Reliability:** Improved performance and availability.



Comparison of Cloud Deployment Models

Model	Advantages	Disadvantages
Public	• Lower costs compared to private and hybrid clouds. • No need for maintenance as it is managed by the provider. • Highly scalable and flexible. • Accessible from anywhere with an internet connection. • Rapid deployment of services.	• Less secure as resources are shared. • Performance depends on internet speed. • Data is controlled by the service provider. • Potential compliance and regulatory issues. • Risk of vendor lock-in.
Private	• High security and privacy. • Better performance and space capacity. • Full control over the cloud environment. • Suitable for organizations with strict compliance needs.	• Higher costs for setup and maintenance. • Requires skilled personnel to manage. • Limited scalability compared to public cloud. • Slower deployment timelines. • Risk of outdated technology.
Hybrid	• Greater security than public cloud. • Flexible resource allocation. • Reduced risk with a mix of private and public clouds. • Better business continuity and disaster recovery	• Complex management. • Dependence on service providers. • Potential data integration challenges. • Higher costs for managing multiple environments

Comparison of Cloud Deployment Models

Model	Advantages	Disadvantages
Community	• Cost-effective for the community. • Better security than public cloud. • Encourages collaboration and knowledge sharing. • Customizable to meet community needs.	• Not suitable for every organization. • Security not as high as private cloud. • Limited scalability and performance. • Potential conflicts among community members.
Multi-cloud	• Avoids dependency on a single vendor. • Enhances reliability and resilience. • Optimizes costs and services. • Meets compliance requirements. • Access to specialized services.	• Increased management complexity. • Potential higher costs. • Challenges in data governance. • Integration and compatibility issues.

Challenges in Cloud Computing

- Large computing infrastructure and energy needs for the Cloud Service Providers(CSPs)
- Cost optimization
- High dependence on network
- Multi-cloud environments and lack of knowledge and expertise
- Issues and challenges concerning the integration of real and virtual infrastructure across geographies, such as security and legislation
- Security challenges in terms of confidentiality, secrecy, and protection of data in a cloud environment since organisations do not own the infrastructure they use to process security, data and to store information.
- Legal issues since the computing infrastructure can exist across diverse geographical locations with dynamic legal and political environments.
- High dependency on Service Providers – Vendor lock-in
- Complexity & Skill Gaps : Cloud-native services evolve quickly!
- Downtime & Reliability